

# CNC Cell Maintenance Manual

Synthetic reference document - AI Factory Intelligence Command Center. For demonstration only; not a real operating procedure.

## 1.0 Machine Types and Duty Ratings

Type L cells are rated for light-duty batch work and tolerate the widest process variation. Type M cells are the general-purpose workhorse. Type H cells run tight-tolerance parts and have the lowest allowable overstrain margin. Maintenance intervals scale with duty: L every 500 machine-hours, M every 350, H every 250.

## 2.0 Preventive Maintenance Schedule

2.1 Daily: check coolant level and concentration, inspect way covers, confirm air temperature is within 295 - 305 K and process temperature is no more than 12 K above air temperature.

2.2 Weekly: lubricate ball screws, verify spindle runout, download the controller fault log.

2.3 At each PM interval: replace spindle coolant filter, check drive belt tension, calibrate the torque sensor per the Calibration Checklist.

## 3.0 Tool Wear Management (TWF)

3.1 Tool wear is tracked in cumulative minutes of cutting time. The wear life for standard carbide tooling is 200 minutes nominal, 240 minutes absolute maximum.

3.2 Log every tool change with the tool ID and the wear reading at removal. A tool removed below 150 minutes indicates a feeds-and-speeds problem to investigate.

3.3 Worn tools raise cutting torque, which compounds overstrain risk on Type H cells. Prioritise tool changes there.

## 4.0 Thermal System and Heat Dissipation (HDF)

4.1 The chiller must hold the process-to-air temperature difference at or above 8.6 K under load. A shrinking difference with rotational speed below 1380 rpm is the classic HDF precursor.

4.2 Clean the heat exchanger monthly. A fouled exchanger is the most common root cause of repeat HDF alerts.

4.3 If HDF alerts repeat after cleaning, check the coolant pump impeller and the thermostatic expansion valve.

## 5.0 Drivetrain, Power and Overstrain (PWF, OSF)

5.1 Mechanical power should sit between 3.5 and 9 kW during a normal cut. Readings outside that band trip the PWF logic. Low power with rising torque means the spindle is bogging down - reduce depth of cut.

5.2 Overstrain (OSF) is driven by tool wear times torque. Keep the product under the Type threshold (L 11000, M 12000, H 13000 min\*Nm).

5.3 After any OSF event, inspect the spindle bearings and the axis couplings for backlash before returning the machine to service.

## 6.0 Corrective Maintenance Workflow

6.1 Triage the alert against sensor history in the Command Center. Confirm the failure mode (TWF/HDF/PWF/OSF/RNF).

6.2 Raise a work order with urgency LOW, MEDIUM or HIGH. HIGH is reserved for conditions that can cause tool breakage, thermal damage or an unplanned stop within one shift.

6.3 Perform LOTO per the Safety SOP section 2.0. Complete the repair, update the maintenance log, and run a 15-minute verification cut before release.

## **7.0 Spare Parts and Consumables**

Keep on hand: carbide insert sets for each cell, two spindle coolant filters per machine, one spare drive belt per Type, and a calibrated reference torque cell. Reorder when stock falls below 30 percent.

## **8.0 Record Keeping**

Every intervention must produce a dated record containing the machine ID, failure mode, parts used, technician, approver, and the post-repair verification result. Records feed the reliability model and must not be skipped.